Casimir force between metal plate and dielectric plate*

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Received April 30, 1999

Abstract The Casimir effect between metal plate and dielectric plate is discussed with 1 + 1-dimensional potential model without using cut-off method. Calculation shows that the Casimir force between metal plate and dielectric plate is determined not only by the potential V_0, the dielectric thickness and the distance a between the metal plate and dielectric plate, but also by the dimension of the vessel. When a is far less than the dimension of the vessel, the Casimir force $F_c \propto a^{-1}$; conversely $F_c \propto a^{-2}$. This result is significant for Casimir force experiment.

Keywords: Casimir effect, renormalization, specific calculation.

Casimir predicted that there is an attractive force between two neutral conducting plates, namely the so-called Casimir force[1] 50 years ago. This interaction comes from the zero-point fluctuations. Casimir force widely appears in physics, since the fluctuations are universal statistical phenomena. The Casimir forces in different cases have been calculated by many authors[1—7]. There are two main methods for the calculation. One is quantum fields theory with finite boundary, and the other is fluctuation theory or its equivalence. Casimir force[3] has been proven to exist by many experiments, but only one experiment is quantitatively valid so far which uses a torsion pendulum for measurement at 5% level. Evidently, it is still necessary to determine the various properties of the Casimir force by means of precision measurement. To this end, the theoretical research should provide more guidance for experiments.

In quantum field theory, Casimir effect can be interpreted as the interaction between quantized field and external classical field. The exact calculation is possible only in some special cases, which can replace external fields by ideal symmetrical boundary conditions. To deal with the nonideal conductor or dielectric, a finite potential model can be used[3,4]. One can also introduce the reflection and transmission coefficients of the dielectric for discussion[9—11]. But, so far, there has been no experiment to accurately verify the Casimir force between nonideal metal plates. In detecting the Casimir force between the nonideal conductors or dielectrics, we should consider the existence of walls of metal vessel, which will reflect the penetrative waves. In this paper, we calculate the Casimir force between metal plate A (ideal conductor) and dielectric plate B, using 1 + 1-dimensional potential model. One wall of the metal vessel is described by the metal plate C as shown in fig. 1. Here, we take the entire space as $a + b + c$, and usually $a + b + c \gg a$. Theoretical analysis shows that the expression of the Casimir force in this case is convergent without using cut-off function.

* Project supported by the Plastic Shape Simulateion & Mould Technology Laboratory (98-8) and the National Natural Science Foundation of China (Grant Nos. 19425008 and 19835040).

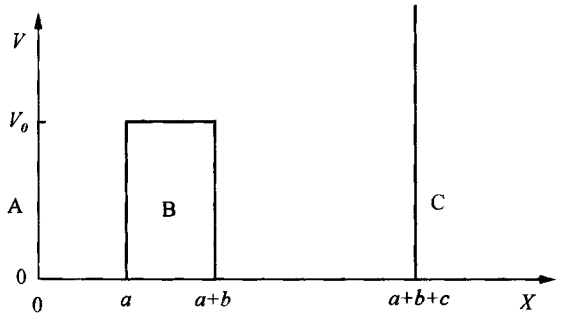


Fig. 1. Potential model of the Casimir effect between metal plate and dielectric plate where $x < 0$ is the metal plate A. $a < x < a + b$ is the dielectric plate B. a, the distance between metal plate and dielectric plate; b, the thickness of the dielectric. $x > a + b + c$ represents one wall of the vacuum vessel.

Casimir force between dielectric plates in 1 + 1-dimensional version has been studied through introducing the reflection and transmission coefficients by some authors[9—11]. Because these two coefficients depend on the frequency seriously, it is difficult to test the result obtained theoretically. Using the finite potential model of the dielectric, we can study the influence of the parameters of the potential height and thickness, as well as the dimension of the vessel. The calculation shows that the Casimir force F_c between metal and dielectric plates is complicatedly dependent on the distance due to the existence of the vacuum vessel. At large distance, $F_c \propto a^{-2}$ and at small distance $F_c \propto a^{-1}$. Moreover, we show the existence of the invariance of the ratio $e(\lambda, a)$ under the scale transformation $(\lambda, a) \to (\lambda/t, at)$ for any real t when the distance a is small and potential $V_0(x) = \lambda\delta(x)$. All calculations in this paper are dimensionless.

1 Theory calculation in 1 + 1-dimension model

A metal is a dissipative system. It is difficult to deal with the Casimir force in this system. An approximate method often used when treating the penetration of electromagnetic waves in dielectric is the finite potential model[3,4]. In the 1 + 1-dimensional model, the vacuum fluctuations of the two modes of the electromagnetic field can be obtained by solving the massless one-dimension Klein-Gordon equation

$$\partial_t^2\varphi - \Delta\varphi + V(x)\varphi = 0, \tag{1}$$

where $\hbar = c = 1$. The scalar potential $V(x)$ in eq. (1) is written as (fig. 1)

$$V(x) = \begin{cases} \infty, & x \leqslant 0, \\ 0, & 0 < x \leqslant a, \\ V_0, & a < x \leqslant a + b, \\ 0, & a + b < x \leqslant a + b + c, \\ \infty, & x > a + b + c. \end{cases} \tag{2}$$

The solutions of eq. (1) can be discussed in two cases, and the orthogonal wave functions have the forms

$$\varphi_n^{\pm}(x, t) = \frac{1}{\sqrt{(a + b + c)\omega_n}} \exp(\pm i\omega_n t)\, \chi_n(x).$$

When $\omega_n^2 = k_n^2 < V_0$,

$$\chi_n(x) = \begin{cases} A\sin k_n x, & 0 < x \leqslant a, \\ B_1 e^{\mu_n x} + B_2 e^{-\mu_n x}, & a < x \leqslant a + b, \\ C\sin k_n(a + b + c - x), & a + b < x \leqslant a + b + c, \end{cases} \tag{3}$$

where $\mu_n = \sqrt{V_0 - k_n^2}$. The wave functions could be orthonormalized by the indefinite scalar product,

$$(\varphi_n^{\pm}, \varphi_m^{\pm}) = \mp \delta_{nm}, (\varphi_n^{\pm}, \varphi_m^{\pm}) = 0, \tag{4}$$

where the normalized brackets are defined as

$$(f, g) = \int (f^* \, i\partial_t g - g^* \, i\partial_t f) \mathrm{d}x.$$

The discrete spectrum $\omega_n = k_n$ is determined by the continuity of the functions and their derivatives at $x = a$ and $a + b$, i.e.

$$\begin{vmatrix} (\mu_n \sin k_n a + k_n \cos k_n a) \mathrm{e}^{\mu_n b} & \mu_n \sin k_n c - k_n \cos k_n c \\ (\mu_n \sin k_n a - k_n \cos k_n a) \mathrm{e}^{-\mu_n b} & \mu_n \sin k_n c + k_n \cos k_n c \end{vmatrix} = 0. \tag{5}$$

The relationships of A, B_1, B_2, C, and k_n could be also determined correspondingly,

$$B_1 = \frac{1}{2} A \left(\sin k_n a + \frac{k_n}{\mu_n} \cos k_n a \right) \mathrm{e}^{-\mu_n b}, \tag{6}$$

$$B_2 = \frac{1}{2} A \left(\sin k_n a - \frac{k_n}{\mu_n} \cos k_n a \right) \mathrm{e}^{\mu_n b}, \tag{7}$$

$$C^2 = \frac{(\mu_n \sin k_n a)^2 - (k_n \cos k_n a)^2}{(\mu_n \sin k_n c)^2 - (k_n \cos k_n c)^2} A^2. \tag{8}$$

Using orthonormalized condition (4), we have

$$\frac{1}{a + b + c} \left[A^2 \left(a - \frac{\sin 2k_n a}{2k_n} \right) + C^2 \left(c - \frac{\sin 2k_n c}{2k_n} \right) + A^2 \left[\sin^2 k_n a - \left(\frac{k_n}{\mu_n} \cos k_n a \right)^2 \right] b \right]$$

$$+ \frac{1}{a + b + c} \left[\frac{1}{4} A^2 \left(\sin k_n a + \frac{k_n}{\mu_n} \cos k_n a \right)^2 \frac{\mathrm{e}^{-2\mu_n b} - 1}{\mu_n} + \frac{1}{4} A^2 \left(\sin k_n a + \frac{k_n}{\mu_n} \cos k_n a \right)^2 \frac{\mathrm{e}^{2\mu_n b} - 1}{\mu_n} \right] = 1. \tag{9}$$

By combining eq. (9) with eqs. (6)—(8), the coefficients A, B_1, B_2 and C can be determined.

When $k_n^2 = \omega_n^2 > V_0$,

$$\chi_n(x) = \begin{cases} A \sin k_n x, & 0 < x \leqslant a, \\ B_1 \sin \mu_n x + B_2 \cos \mu_n x, & a < x \leqslant a + b, \\ C \sin k_n (a + b + c - x), & a + b < x \leqslant a + b + c, \end{cases} \tag{10}$$

where $\mu_n = \sqrt{k_n^2 - V_0}$. In the same way A, B_1, B_2 and C can be determined and $k_n = \omega_n$ satisfies the condition

$$k_n^2 \sin \mu_n b \cos k_n (a + c) + V_0 \sin k_n a \sin k_n c \sin \mu_n b + k_n \mu_n \cos \mu_n b \sin k_n (a + c) = 0. \tag{11}$$

The energy density of the coupled modes in eqs. (3) and (10) in the field φ is

$$T_{00} = \frac{1}{2} (| \partial_t \varphi |^2 + | \partial_x \varphi |^2 + V \varphi^2). \tag{12}$$

For the modes in eq. (3)

$$T_{00}(x) = \frac{1}{D} \begin{cases} \omega_n^2 A^2, & x \in (0, a], \\ \mu_n^2 (B_1 \mathrm{e}^{\mu_n x} - B_2 \mathrm{e}^{-\mu_n x})^2 + (\omega_n^2 + V_0)(B_1 \mathrm{e}^{\mu_n x} + B_2 \mathrm{e}^{-\mu_n x})^2, & x \in (a, a + b], \\ \omega_n^2 C^2, & x \in (a + b, a + b + c], \end{cases} \tag{13}$$

where $D = 2(a + b + c)\omega_n$. The total energy has a simple form

$$E_n = \int_0^{a+b+c} T_{00}(x) \mathrm{d}x$$

$$= \frac{1}{2}\omega_n \frac{1}{a+b+c}\left[A^2 a + C^2 c + \frac{V_0}{\omega_n^2}\int (2B_1^2 e^{2\mu_n x} + 2B_2^2 e^{-2\mu_n x})dx + 4B_1 B_2 b\right] = \frac{1}{2}\omega_n. \quad (14)$$

Analogous result is obtained for the modes in eq. (10), $E_n = \omega_n/2$.

Any solution of eq. (1) can be expanded in $\varphi_n^{\pm}$ as

$$\varphi(x,t) = \sum_n [a_n^- \varphi_n^-(x,t) + a_n^+ \varphi_n^+(x,t)], \quad (15)$$

and the coefficients are determined by the orthonormalized conditions of eq. (4). In the occupation-number representation, the amplitudes $a_n^{\pm}$ are creation and annihilation operators, so the vacuum energy without renormalization is the expectation value $\langle 0|T_{00}|0\rangle$. Then the total energy for the vacuum state is

$$E = \int_0^{a+b+c} \langle 0|T_{00}|0\rangle dx = \frac{1}{2}\sum_{n=1}^{\infty} \omega_n. \quad (16)$$

This summation is infinite since the frequencies ω_n increase without boundary in the limit $n \to \infty$. Setting $V_0 = 0$ ($\omega_n = n\pi/(a+b+c)$) and using zeta function regularization method or cut-off method, we will get the Casimir energy for two metal plates separated by $a+b+c$. Here we want to calculate the Casimir force between a metal plate A and a dielectric plate B with a finite potential V_0 under the influence of the other wall of the vacuum vessel (metal plate C) as shown in fig. 1. Since the dimension of the vessel is a constant, we can set $a+b+c = L$ (constant). Therefore, Casimir force between plates A and B has the form:

$$F_c(V_0, a, b, c) = \frac{1}{2}\sum_n \frac{\partial \omega_n}{\partial a} = \frac{1}{2}\sum_n \frac{\partial k_n}{\partial a}, \quad (17)$$

where $c = L - a - b$, b is a constant, and $\partial k_n/\partial a$ is determined by eqs. (5) and (11). The number of the low frequency modes of series $\partial k_n/\partial a$ determined by eq. (5) is finite. So it is only necessary to consider the high frequency modes of the series determined by eq. (11) to prove the convergence of summation of the series.

In eq. (11), for any finite potential V_0, when $n \to \infty$, we have discrete spectrum $k_n = (n + \delta_n)\pi/(a+b+c)$, where δ_n can be written in the form of a series solution

$$\delta_n = \frac{a+b+c}{2n\pi^2} V_0 b - (-1)^n \frac{(a+b+c)^2}{2n^2\pi^3} V_0 \cos\left(n\pi \frac{c-a}{a+b+c}\right)\sin\left(n\pi \frac{a}{a+b+c}\right) + O\cdot\left(\frac{1}{n^3}\right), \quad (18)$$

If $n \to \infty$, then $\delta_n \to 0$. According to eqs. (11) and (18), $\partial k_n/\partial a$ can also be written as

$$\begin{aligned}\frac{\partial k_n}{\partial a} &= (-1)^{n+1}\frac{V_0}{n\pi}\sin\left(n\pi\frac{c-a}{a+b+c}\right)\sin\left(n\pi\frac{b}{a+b+c}\right) \\ &\quad + (-1)^n \frac{V_0^2 b(a+b+c)}{2n^2\pi^2}\sin\left(n\pi\frac{c-a}{a+b+c}\right)\cos\left(n\pi\frac{b}{a+b+c}\right) + O\cdot\left(\frac{1}{n^3}\right) \\ &= (-1)^{n+1}\frac{V_0}{2n\pi}\left[\cos\left(n\pi\frac{c-a-b}{a+b+c}\right) - \cos\left(n\pi\frac{c-a+b}{a+b+c}\right)\right] \\ &\quad + (-1)^n \frac{V_0^2 b(a+b+c)}{4n^2\pi^2}\left[\sin\left(n\pi\frac{c-a-b}{a+b+c}\right) + \sin\left(n\pi\frac{c-a+b}{a+b+c}\right)\right] + O\cdot\left(\frac{1}{n^3}\right). \quad (19)\end{aligned}$$

By means of formula $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\cos(nx)}{n} = \ln\left(2\cos\frac{x}{2}\right)$, $-\pi < x < \pi$, the convergence of the

summation $\sum_{n=1}^{\infty} \frac{1}{2} \frac{\partial k_n}{\partial a}$ can be proved easily without using cut-off function.

2 Numerical evaluation

2.1 F_c versus V_0

The Casimir force between a metal plate and a dielectric plate is given by eq. (17). In the experiment, $a + b + c \gg a$ is usually satisfied. The ratio of the Casimir force between the plates A and B to that between the ideal conducting plates separated by a is defined as

$$e(V_0, a, b, c) = \frac{F_c(V_0, a, b, c)}{-\pi/(24a^2)}, \tag{20}$$

where $-\pi/(24a^2)$ is the Casimir force between two ideal conducting plates separated by a. According to eqs. (5) and (11), we have $e(V_0, a, b, c) = e(V_0/t^2, ta, tb, tc)$ for any $t > 0$. This means the scale transformation invariance of $e(V_0, a, b, c)$. So we can assume $a + b + c = 1$ in technique. For convenience of numerical evaluation, we replace eq. (17) by the following equation:

$$F_c(V_0, a, b, c) = \lim_{\substack{\Delta a \to 0 \\ M \to \infty}} \frac{1}{2} \frac{\sum_{n=1}^{M} \omega_n\left(V_0, a + \frac{\Delta a}{2}, b, c - \frac{\Delta a}{2}\right) - \sum_{n=1}^{M} \omega_n\left(V_0, a - \frac{\Delta a}{2}, b, c + \frac{\Delta a}{2}\right)}{\Delta a}, \tag{21}$$

where $a + b + c = L = 1$, $\omega_n(V_0, a, b, c)$ is determined by eqs. (5) and (11). The choice of the parameters M and Δa is determined by the potential height V_0 and the calculation precision required.

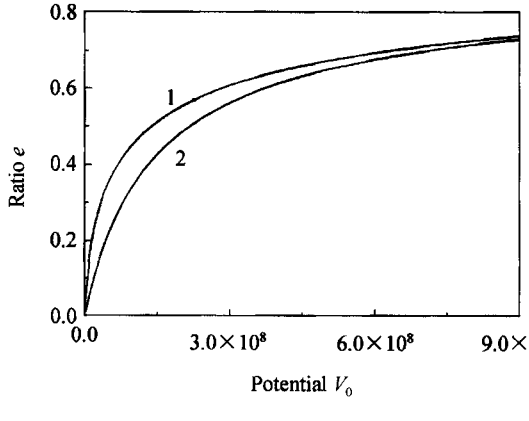


Fig. 2. Dependence of the ratio $e(V_0, a, b, c)$ on the potential V_0. $a = 0.000\ 2$, $a + b + c = 1$. Curves 1 and 2 correspond to $b = 0.6$, $b = 0.000\ 06$, respectively.

Curve 1 in fig. 2 shows the relationship of $e(V_0, a, b, c)$-V_0 for $a = 0.000\ 2$, $b = 0.6$ and $c = 0.399\ 8$. In practical experiment, one can choose $a = 1$ μm, $b = 3$ mm, and $c = 2$ mm with the help of the auxiliary installation. It is reasonable to set $a = 0.000\ 2$, $b = 0.6$, $c = 0.399\ 8$ in the same proportional relation. Curve 2 in fig. 2 represents the relationship of $e(V_0, a, b, c)$-V_0 for $a = 0.000\ 2$, $b = 0.000\ 06$ and $c = 0.999\ 74$. Numerical calculation shows that the parameter b has a distinct influence on the $e(V_0, a, b, c)$ only when b is close to or less than a. With the same a, the curve for $b = 0.000\ 6$ almost overlaps curve 1 in fig. 2. In addition, one can expect $e(V_0, a, b, c) \to 1$ as $V_0 \to \infty$ for fixed a and b.

For the 3 + 1-dimensional model, $e(V_0, a, b, c)$ could be introduced in the same way, which is measurable realistically in the experiments.

2.2 F_c versus a

Due to the restriction of the entire space $a + b + c = 1$, the Casimir force is in a complicated way dependent on the distance (fig. 3). For given V_0 and b, $e(V_0, a, b, c)$ decreases as a decreases. When $V_0 = 10^8$, $a > 0.001$, $F_c \propto a^{-2}$ and $F_c \propto a^{-1}$ if $a < 0.001$.

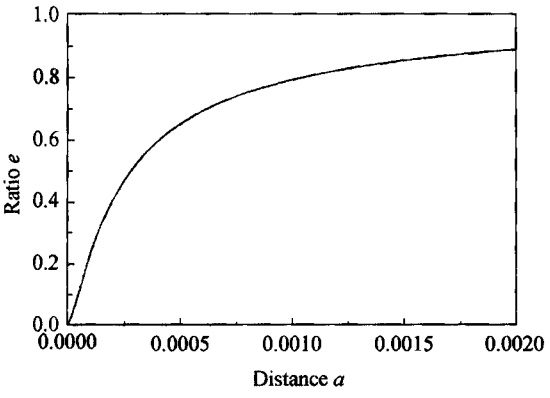


Fig. 3. Dependence of the ratio $e(V_0, a, b, c)$ on distance a, $V_0 = 10^8$, $b = 0.000\ 1$ and $a + b + c = 1$.

any real t when $a \ll 1$ (here $a + c = 1$). a is inversely proportional to λ for a given $e(\lambda, a)$.

3 Physical explanation

Detecting Casimir force is done in a vacuum vessel. We should consider the influence of the potential and the dimension of the vacuum vessel because of the penetration of electromagnetic waves in the dielectric. According to quantum fields theory, the Casimir force is dependent on the geometry and topology of the space. In the case of two ideal conducting plates separated by a, the regions inside and outside the plates are separated. The Casimir force is only simply dependent on a. But in our model, the regions inside and outside get together, since the potential is finite. Therefore, the Casimir force is related with the dimension of the vessel.

It is noted that if we set $b \to 0$ and $V(x) = \lambda\delta(x - a), x \in (0, a + c)$, then $e(\lambda, a) \to 1$ as $\lambda \to \infty$, $e(\lambda, a) \to 0$ as $a \to 0$.

Figure 4 tells us that $e(\lambda, a) \to 0$ as $a \to 0$, which means the penetration of potential is increased between the metal and the dielectric plates when the a decreases. Curve 1 is $e(\lambda, a = 0.01)$. Curves 2, 3 and 4 correspond to $a = 0.001$, $0.000\ 1$ and $0.000\ 01$, respectively. Numerical evalution shows that curves 2, 3 and 4 are 10, 100 and 1000 times of curve 1 in the abscissa, respectively. So $e(\lambda, a)$ is invariant under the scale transformation $(\lambda, a) \to (\lambda/t, at)$ for

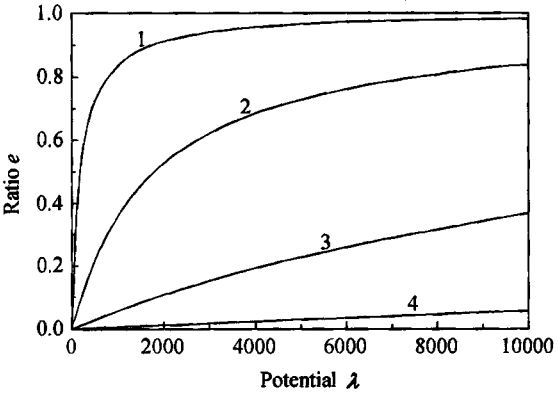


Fig. 4. Dependence of the ratio $e(\lambda, a)$ on the potential λ. Curves 1, 2, 3 and 4 correspond to $a = 0.01$, 0.001, $0.000\ 1$, and $0.000\ 01$, respectively.

In eq. (2) when V_0 is very large, $a + b + c \gg a$ and $b \to 0$ (as δ potential), the energy in $0 < x < a$ is $E_a \approx \frac{1}{2}\sum_n \frac{n\pi}{a}$, and the energy in space $a + b < x < a + b + c$ is $E_c \approx \frac{1}{2}\sum_n \frac{n\pi}{c}$. So the total energy is $E = E_a + E_c$. When a decreases to $a - \Delta a$, the model $n\pi/a$ in the space $0 < x < a$ increases to $n\pi/(a - \Delta a)$ and then $E_{a-\Delta a} \approx E_a(1 + \Delta a/a)$, while the model $n\pi/c$ in the space $a + b < x < a + b + c$ decreases to $n\pi/(c + \Delta a)$ and $E_{c+\Delta a} \approx E_c(1 - \Delta a/c)$. Since the energy density in $0 < x < a$ is c/a times that in $a + b < x < a + b + c$, the total

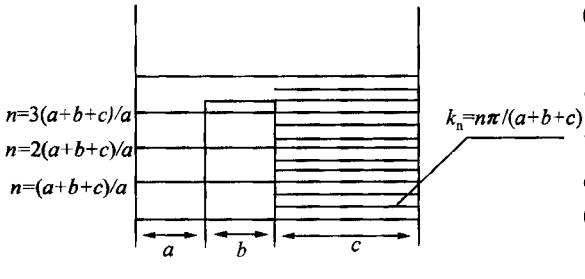


Fig. 5. From the entire space, all models are near $n\pi/(a + b + c)$. The unequilibrium between the models n satisfying $n \approx i(a + b + c)/a$ $(i = 1, 2, 3, \cdots)$ with the other models forms the Casimir force.

energy in the entire space still remains constant approximately. For any finite potential V_0, the lower models of $\omega_n^2 < V_0$ are influenced evidently. Numerical calculation shows that the total energy for the entire space will decrease a little as the two plates are close to each other. That forms the Casimir force (fig. 5). For a given V_0, the $e(V_0, a, b, c)$ will decrease as a decreases and $e(V_0, a, b, c) \to 0$ as $a \to 0$. The reason is that when a decreases to 0, the model $\omega_n^2 < V_0$ will increase to $\omega_n^2 > V_0$. Their contribution to the Casimir effect $e(V_0, a, b, c)$ will decrease. But the model in space $a + b < x < a + b + c$ scarcely changes, since $c \gg a$.

Under the influence of the dimension of the vessel, the Casimir force will be proportional to a^{-1} when the separation between the conducting plate and the dielectric one is very small. This conclusion is correct for the 3-dimensional case.

Acknowledgment The authors thank M.L.Ge and Liu Liao for valuable discussion.

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